Research on the Impact of Gamification on Brand Equity

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ABSTRACT

This paper aims to clarify the relationship between gamification and brand equity based on self-determination theory and the stimulus-organism-response model of environmental psychology. It uses a questionnaire to collect data and hierarchical regression analysis to test the hypotheses. This research provides an empirical analysis of how gamification affects brand equity. Research results show that gamification and its three dimensions positively affect brand equity; game dynamics and components positively influence consumers' autonomy, competence, and relatedness, while game mechanics have no significant effect on the three basic needs. Autonomy, competence, and relatedness positively impact brand equity and partially mediate the relationship between gamification and brand equity. Openness, one of the five core personalities, positively moderates the impact of autonomy and relatedness on brand equity and negatively moderates the impact of competence on brand equity. This paper not only enriches self-determination theory and gamification theory but also provides suggestions for gamification practice.

CCS CONCEPTS

• **: Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI.

KEYWORDS

Gamification, Brand equity, Self-determination theory, Openness

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1 INTRODUCTION

With the globalization of business and the digitization of information, the value of brand equity continues to come to the fore. In an increasingly competitive market, how to increase the value of brands has become the main focus for companies. At the same time, the development of information technology and social media has led to the growth of the gaming industry, and the use of games to drive economic development has become a new development trend. More and more companies strengthen their connection with consumers through gamification to improve brand experience and enhance the brand value. Alipay's Ant Forest is one of the typical gamification practices. Alipay is one of the world's most popular online payment and lifestyle platforms. In August 2016, it launched Ant Forest, which rewards users with 'green energy points' every time they take a step to reduce their emissions, such as cycling to work, going paperless, and buying sustainable products. These green energy points grow into a virtual tree, which Alipay matches by planting a real tree or protecting a protected area. It attempts to continuously raise users' awareness of environmental protection by setting goals (creating a virtual tree) and interacting with friends (watering and collecting energy for each other's trees). Pinduoduo, a third-party social e-commerce platform, is another successful example of gamification. Marketing activities such as group buying, inviting friends to participate in price-cutting, and lotteries are essentially a game. Consumers get different experiences from these different "games," such as the money-saving experience of group buying, the social experience of inviting, and the surprise experience of lotteries.

Research on gamification marketing is still at the beginning. Previous studies have shown that gamification positively affects brand attitudes [1], brand engagement [2], brand involvement [3], and brand love [4]; fewer studies have dealt with brand equity. This paper constructs a research model based on self-determination theory, focuses on the influence of gamification on brand equity and the influence mechanism, and explores the moderating role of openness in the Big Five personality. The research results of this paper will further reveal the relationship between gamification and brand equity, explore the influence mechanism and boundary conditions of gamification on brand equity, which will have important theoretical value and practical implications.

2 LITERATURE REVIEW

2.1 Gamification

The term gamification originated in the digital media industry. Gamification is the application of game design elements in non-game contexts [5]. From a service marketing perspective, it is a process [6] that enhances services through gamified experiences to support the overall value creation for the user [7]. There are different classifications of gamification elements. Gamification elements involve both extrinsic and intrinsic gamification elements [8]. From top to bottom, the gamification pyramid theory considers the elements of gamification as dynamics, mechanics, and components [9]. The

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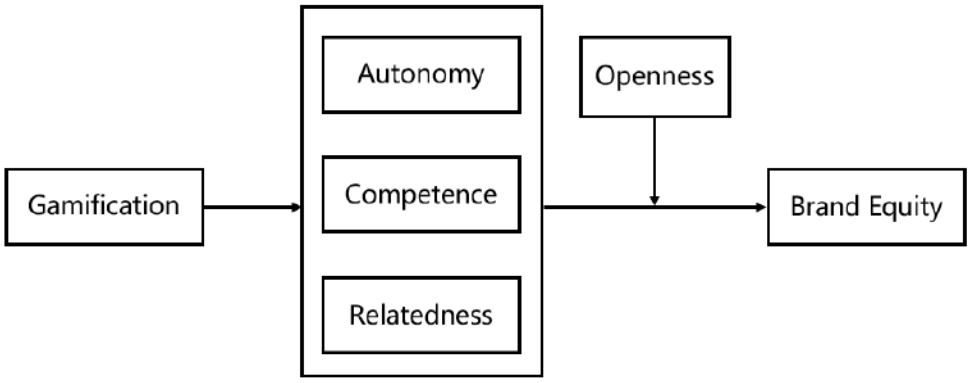


Figure 1: Theoretical Model

main objectives of gamification are in line with the three core marketing concepts of customer stickiness, brand loyalty, and brand awareness [10]. Gamification successfully drives consumers to engage in gamified marketing activities to enhance the value of the experience and increase purchase intention [11].

2.2 Brand equity

Brand equity can be defined from three different perspectives: finance, market and consumer. Finance-based brand equity focuses on cash flow, surplus, and costs. Market-based brand equity is concerned with a brand's performance in the marketplace and emphasizes brand growth and expansion. In this paper, we study customer-based brand equity. Customer-based brand equity is defined as the differential effect of brand knowledge on consumer response to the marketing of the brand [12]. Customer-based brand equity occurs when the consumer is familiar with the brand and has some favorable, strong, and unique brand associations in memory. There are different viewpoints about the dimensions of brand equity. Researchers try to describe and measure brand equity from various aspects. It is a two-dimensional construct including brand awareness and brand image [12],a three-dimensional construct consisting of brand loyalty, perceived quality, and brand association [13],and a four-dimensional construct containing brand association, brand awareness, brand perception, and brand resonance [14], a five-dimensional construct, including brand loyalty, brand awareness, perceived quality, brand association, and other proprietary brand assets (e.g., patents, trademarks, channel relationships, etc.) [15]. This paper will measure brand equity in three dimensions: brand loyalty, brand perception, and brand association. Brand loyalty refers to the degree of consumer preference for a brand, reflecting the degree of trust and reliance on that brand; Brand perception is what customers believe a product or service represents. Brand association is a mental connection between a brand and a concept that a consumer recalls when he comes across that brand.

Brand relationships can promote the brand citizenship behavior of franchisees and ultimately enhance brand equity [16]. Brand experience value significantly impacts brand equity [17]. There is a link between brand personality and sportswear brand equity; four personality dimensions, namely competence, attractiveness, sincerity, and innovation, directly impact brand equity [18].

3 THEORETICAL MODEL AND RESEARCH HYPOTHESIS

3.1 Theoretical Model

This paper constructs a theoretical model based on the Stimulus-Organism-Response (SOR) research paradigm of environmental psychology. The SOR model assumes that stimuli from the external environment trigger a sensory or behavioral response in the organism. The paper argues that gamification is an external stimulus for online consumers. The three psychological needs based on self-determination theory influence brand equity as consumers' psychological response mechanism, and openness in the Big Five personality moderates the effect between three psychological needs and brand equity. The research framework is shown in Figure 1

3.2 Hypothesis Development

3.2.1 The impact of gamification on brand equity. Gamification can facilitate customer engagement, and gamification elements such as challenges, tasks, and rewards can influence customer engagement in both emotional and behavioral dimensions [19]. Gamification is a potential branding tool that helps understand consumer behavior and provides an interactive platform conducive to brand engagement [3]. Gamification positively influences brand awareness and brand loyalty [10]. Gamification positively affects brand engagement and thus increases brand equity [20]. Thus, gamification can impact brand equity, either directly or indirectly.

The gamification elements have different classifications, among which the gamification pyramid theory is the most widely used[9].In the gamification pyramid model, game dynamics is at the highest level, and its elements are used to enhance consumer feelings and emotions, such as emotions, relationships, and growth. Game mechanics are the fundamental processes that drive the gamification process and user participation, such as competition and cooperation, exploration, and resource acquisition. On the other hand, game components are the concrete expressions of game dynamics and mechanics, such as points, badges, and a leaderboard.

Accordingly, we hypothesize:
H1: Gamification positively influences brand equity
H1a: Game dynamics positively influence brand equity
H1b: Game mechanics positively influence brand equity
H1c: Game components positively influence brand equity

3.2.2 The impact of gamification on consumers' psychological needs. In the 1980s, American psychologists Deci and Ryan proposed the self-determination theory [21]. Self-determination theory asserts that people must have their basic needs of autonomy, competence and relatedness met to maintain their physical and mental health [22]. Autonomy refers to the need for individuals to be able to make their own decisions when engaging in activities to achieve the desired goals; competence represents the desire to experience confidence in personal skills and promotes a sense of accomplishment; relatedness means the desire to receive external care and support and the wish to establish mutual respect and connection with others [21].

Games satisfy the basic psychological needs of players. The application of game design elements in non-game contexts does not change the nature of the game and still has an impact on the three basic psychological needs mentioned in self-determination theory [23]. When individuals have the power to make their own choices or have many choices, the sensitivity to autonomy increases, and the willingness to participate in the activity increases accordingly. Conversely, when they are forced to participate in an activity because of external pressure, it is hard to satisfy their autonomy [24]. If the games are highly challenging which will likely inspire gamers but may also create frustration. If gamers feel frustrated, their competence cannot be met [25]. Video games that emphasized different game elements have different impacts on players' needs. Games that emphasized flexible rules led to feelings of competence while games that emphasized social elements led to feelings of relatedness [26].There is social embeddedness of social online gamer and gaming-related ties might be very useful. Enhanced social online gaming time increases the probability of finding online friends. Appropriate social interaction can reduce the likelihood of the gamer falling into frustration or depression [27]. Thus, we hypothesize:

- H2: Gamification positively impacts autonomy
- H2a: Game dynamics positively impact autonomy
- H2b: Game mechanics positively impact autonomy
- H2c: Game components positively impact autonomy
- H3: Gamification positively impacts competence
- H3a: Game dynamics positively impact competence
- H3b: Game mechanics positively impact competence
- H3c: Game components positively impact competence
- H4: Gamification positively impacts relatedness
- H4a: Game dynamics positively impact relatedness
- H4b: Game mechanics positively impact relatedness
- H4c: Game components positively impact relatedness

3.2.3 The impact of consumer psychological needs on brand equity. Brand equity has significant commercial value. It is very important for the company to build strong brand equity and brand value, but the natural source of brand equity and brand value comes from consumers [28]. Brand equity is the brand value-added. Under the influence of the objective world and subjective consciousness, consumers experience a complex brand consumption psychological process and generate brand consumption actions, forming a brand-specific value, that is, brand equity in the minds of consumers [29]. Different types of social media services can satisfy different psychological needs. Furthermore, satisfying basic human needs can drive consumption [30]. The perceived psychological benefits in the brand co-creation process facilitate the establishment of brand co-creation engagement that leads to a strengthened brand relationship [31]. Positive relationships and transcendent customer experiences can strengthen customers' ties to a brand community, delivering a particularly strong brand loyalty [32]. It reveals that the satisfaction of customers' psychological needs can positively impact brand equity. Therefore, we hypothesize:

- H5: Autonomy positively affects brand equity
- H6: Competence positively affects brand equity
- H7: Relatedness positively affects brand equity

3.2.4 Intermediary effect. Based on hypotheses H1-H7, gamification positively affects brand equity, positively affects the three basic needs of autonomy, competence, and relatedness, and the three basic needs positively impact brand equity. According to the above hypothesis, consumers who have their basic psychological needs met through gamified marketing interaction will be more actively involved in brand-related activities, thus cultivating brand loyalty and increasing brand equity. Thus, we hypothesize:

- H8: Autonomy mediates the relationship between gamification and brand equity
- H9: Competence mediates the relationship between gamification and brand equity
- H10: Relatedness mediates the relationship between gamification and brand equity

3.2.5 The moderating role of openness. Openness is an essential personality trait denoting receptivity to new ideas and new experiences. It is one of the five core personalities that drive behavior—known as the five-factor model of personality. An open individual has the characteristics of active imagination, aesthetic sensitivity, concern for inner feelings, preference for diversity, intellectual curiosity, and independent judgment [33]. Open personality trait is positively related to the hedonic value of products and brand influence [34]. Individuals with open personality trait are curious about the internal and external world, open to new ideas and non-traditional values, and sensitive to experiencing positive and negative emotions [33]. Therefore, in gamification marketing, consumers are more likely to perceive emotional changes and generate an emotional response when their psychological needs are met, thus increasing brand equity. Accordingly, we hypothesize:

H11: Openness positively moderates the positive impact of autonomy on brand equity

H12: Openness positively moderates the positive impact of competence on brand equity

H13: Openness positively moderates the positive impact of relatedness on brand equity

Table 1: Means, standard deviations and correlation coefficients

	1	2	3	4	5	6	7	8	9
1. Gender	1								
2.Education	0.035	1							
3.Occupation	-0.243**	-0.265**	1						
4.Gamification	-0.004	0.009	0.101	1					
5. Autonomy	-0.054	-0.052	0.094	0.685**	1				
6.Competence	0.042	0.061	0.063	0.600**	0.564**	1			
7.Relatedness	-0.053	-0.059	0.129	0.605**	0.550**	0.633**	1		
8.Brand Equity	0.001	0.016	0.042	0.689**	0.570**	0.674**	0.636**	1	
9. Openness	-0.034	0.028	0.103	0.508**	0.516**	0.462**	0.543**	0.641**	1
Mean	1.76	2.98	1.51	3.87	4.00	3.73	3.67	3.83	3.84
S.D.	0.42	0.45	1.03	0.52	0.59	0.68	0.66	0.51	0.57

Note: N=187, **p<0.01, *p<0.05

4 EMPIRICAL STUDY

4.1 Descriptive Statistical Analysis and Multicollinearity Test

This paper used a questionnaire survey to collect data and SPSS 25.0 and AMOS 24.0 to test the hypotheses. The randomized sampling method was used and questionnaires were distributed online through Wenjuanxing, a survey platform, to a few universities in Fujian and Anhui provinces, where the sample are located at a moderate level of economic development region. Therefore, the impact of the economic development level on brand equity can be excluded from this research. The sample size is 200, and the final valid sample is 187. Approximately 76% of the sample is university students, with the age groups concentrated in the post-90s and post-00s, accounting for 71% and 28% of the university student sample, respectively. The young people are the leading players in the gaming industry and have strong brand awareness, which indicates the sample representative.

4.1.1 Descriptive statistical analysis. The means, standard deviations, and correlation coefficients between the variables are shown in Table 1. Gamification is positively correlated with brand equity with a correlation coefficient of 0.689, and gamification is actively related to autonomy, competence, and relatedness with correlation coefficients of 0.685, 0.600, and 0.605, respectively; autonomy, competence, and relatedness are directly linked to brand equity with correlation coefficients of 0.570, 0.674, and 0.636, respectively, indicating that the correlation relationships between the main variables are consistent with the expected hypothesis.

4.1.2 Multicollinearity test. A multicollinearity test is performed on the key variables. The SPSS regression results show that the tolerance values for gamification, autonomy, competence, relatedness, and openness are 0.427, 0.453, 0.486, 0.462, and 0.621 respectively (greater than 0.2). The variance inflation factor (VIF) values are 2.345, 2.206, 2.059, 2.163 and 1.610 respectively (less than 5). There is no multicollinearity between the variables.

4.2 Reliability and Validity Analysis

4.2.1 Reliability analysis. This paper used measurement scales published in national and international journals. The variables are evaluated by a five-point Likert scale, ranging from 1-5 to indicate strongly disagree, disagree, neutral, agree, and strongly agree. The measurement scale of gamification is adapted from the gamification pyramid model of Werbach and Hunter [9], and the reliability coefficient of Cronbach's α of gamification as a whole is 0.868, with Cronbach's α of 0.713, 0.715, and 0.724 for game dynamics, game mechanics, and game components respectively. The basic psychological needs measurement scale is adapted from Standage et al. [35], with Cronbach's α of 0.728, 0.800, and 0.690 for autonomy, competence, and relatedness. The brand equity measured from the customer's perspective is adapted from Yoo and Donthu [13]. The overall Cronbach's α of brand equity is 0.894, with Cronbach's α of 0.728, 0.758, and 0.781 for brand perception, association, and loyalty. The measurement scale of openness is adapted from the Big Five personality scale developed by Wang et al. in the Chinese context [36], and Cronbach's α of openness is 0.849. The reliability coefficients of Cronbach's α of all variables in this paper are essentially greater than 0.7, which indicates that all the variables' measurement scales have good reliability.

4.2.2 Validity analysis. The average variance extracted (AVE) comparison method was used to test the discriminant validity between the primary variables. The square root of AVE for each primary variable is greater than the correlation coefficient, which suggests good discriminant validity (see Table 2 for details).

The Kaiser-Meyer-Olkin (KMO) test was performed on the multidimensional constructs, and if the value of KMO is greater than 0.5, the variable is suitable for factor analysis. The values of KMO for game dynamics, game mechanics, and game components in gamification are 0.650, 0.653, and 0.680. the values of KMO for brand perception, brand association, and brand loyalty in brand equity are 0.664, 0.683, and 0.703; all p-values are 0.000 less than 0.05, reaching a significant level, and the variables are suitable for factor analysis.

For the exploratory factor analysis of gamification, the fixed number of extracted factors was three, and the cost method and the maximum variance method were used. The cumulative percentage of variance explained by the three factors of game dynamics,

Table 2: Comparison of correlation coefficients with AVE square root

	Gamification	Autonomy	Competence	Relatedness	Brand Equity	Openness
Gamification	0.689					
Autonomy	0.685	0.679				
Competence	0.600	0.564	0.762			
Relatedness	0.605	0.550	0.633	0.653		
Brand Equity	0.689	0.570	0.674	0.636	0.716	
Openness	0.508	0.516	0.462	0.543	0.641	0.699

Note: The diagonal data are the square root of AVE, and the other data are the correlation coefficients between the variables.

mechanics, and components is 67.2%, indicating that the number of extracted common factors is appropriate. Then, AMOS24.0 was used for confirmatory factor analysis. The chi-square test shows that the chi-square freedom ratio (χ^2/df) is 1.407, less than 3. The factor loading coefficients are essentially greater than 0.5, the value of AVE is 0.4752, and the value of composite reliability (CR) is 0.8897, indicating a good convergent validity.

The exploratory factor analysis of brand equity shows that the cumulative percentage of variance explained by brand loyalty, brand perception and brand association is 67.2%, indicating that the number of extracted common factors is appropriate. The chi-square test shows that the chi-square freedom ratio (χ^2/df) is 1.677, less than 3. The factor loading coefficients are all larger than 0.5, the value of AVE is 0.5124, and the value of CR is 0.9042, indicating an ideal convergent validity.

4.3 Hypothesis Testing

4.3.1 Direct effects test. The paper used hierarchical regression to test the hypotheses. Gamification positively affects brand equity (M4, β =0.690, p<0.01), and hypothesis H1 is supported (see Table 3 for details). The elements of gamification have a positive impact on brand equity, i.e., game dynamics impact brand equity positively (M4, β =0.304, p<0.01), game mechanics impact brand equity positively (M4, β =0.197, p<0.01), and game components impact brand equity positively (M4, β =0.294, p<0.01). Hypotheses H1a, H1b, and H1c are supported (see Table 4 for details).

4.3.2 Intermediary effects test. For the test of mediating effects, the analysis was conducted in four steps, as suggested by Baron and Kenny [37].

Step 1, test the effect of the independent variable on the dependent variable (hypothesis H1, tested).

Step 2 test the effect of independent variables on mediating variables (hypotheses H2, H3, and H4). It was found that gamification positively influences autonomy (M1, β = 0.683, p < 0.01); gamification positively influences competence (M2, β = 0.601, p < 0.01); and gamification positively influences relatedness (M3, β = 0.604, p < 0.01); and hypotheses H2, H3, and H4 are supported by the data (see Table 3 for details).

The different effects of the three dimensions of gamification on consumers' psychological needs are also one of the focuses of this paper. Game dynamics positively affect autonomy, competence, and relatedness (M1, β =0.326, M2, β =0.199, M3, β =0.216, p<0.01), and hypotheses H2a, H3a, and H4a are supported. There is no significant effect of game mechanics on autonomy, competence, and relatedness, and hypotheses H2b, H3b, and H4b are not supported. The game components positively impact the autonomy, competence, and relatedness (M1, β =0.373, M2, β =0.369, M3, β =0.356, p<0.01), and hypotheses H2c, H3c, and H4c are supported. (See Table 4 for details).

Step 3 test the effect of mediating variables on the dependent variable (hypotheses H5, H6, and H7). It was found that autonomy, competence, and relatedness have a positive impact on brand equity (M1, β =0.121, M1, β =0.375, M1, β =0.298, p<0.01), and hypotheses H5, H6, and H7 are supported by the data (see Table 5 for details).

Step 4, test the mediating effects (hypotheses H8, H9, and H10). We noticed that when the mediating variables are introduced into the regression equation, the relationship between the independent and dependent variables remains significant (M5, β =0.556, M6, β =0.445, M7, β =0.475, P<0.01), and the coefficients of the mediating variables are significant (M5, β =0.196, M6, β =0.408, M7, β =0.356, P<0.01). In other words, autonomy, competence, and relatedness partially mediate between gamification and brand equity (see Table 3 for details).

4.3.3 Moderating effects test. The hierarchical regression analysis shows that the coefficient of the interaction item between autonomy and openness is positive (M3, β = 0.152, P < 0.05), and the coefficient of the interaction item between competence and openness is negative (M3, β = -0.217, P < 0.05), and the coefficient of the interaction item between relatedness and openness is positive (M3, β = 0.156, P < 0.05), all of which are significant. Hypotheses H11 and H13 are supported by the data, and the moderating effect of H12 is negative, contrary to the hypothesis (see Table 5 for details).

5 CONCLUSION AND DISCUSSION

5.1 Conclusion

The paper constructs a theoretical model based on self-determination theory and uses the stimulus-organism-response model to investigate the relationship between gamification and brand equity. Research results show that gamification and its three dimensions positively affect brand equity; game dynamics and components positively affect consumers' autonomy, competence, and relatedness, while game mechanics have no significant effect on the three basic needs. Autonomy, competence, and relatedness positively impact brand equity and partially mediate the relationship between gamification and brand equity. Openness, one of the five core personalities, positively moderates the impact of autonomy

Table 3: Hierarchical regression analysis of direct and Intermediary effects

	Autonomy	Competence	Relatedness	Brand Equity			
	M1	M2	M3	M4	M5	M6	M7
Control variables							
Gender	-0.023	0.025	-0.030	-0.013	-0.008	-0.023	-0.002
Age	0.079	-0.05	0.052	-0.047	-0.063	-0.027	-0.066
Education	-0.055	0.053	-0.061	0.009	0.020	-0.013	0.031
Independent variable							
Gamification	0.683**	0.601**	0.604**	0.690**	0.556**	0.445**	0.475**
Mediating variables							
Autonomy					0.196**		
Competence						0.408**	
Relatedness							0.356**
R^2	0.481	0.367	0.375	0.477	0.496	0.582	0.556
ΔR^2	0.467	0.351	0.365	0.453	0.019	0.105	0.079
ΔF	42.170**	26.328**	27.325**	165.411**	7.172**	45.636**	32.192**

Note: N=187, **p<0.01, *p<0.05

Table 4: Hierarchical regression analysis of the three dimensions of gamification

	Autonomy	Competence	Relatedness	Brand Equity
	M1	M2	M3	M4
Control variables				
Gender	-0.022	0.025	-0.31	-0.012
Age	0.070	-0.059	0.044	-0.049
Education	-0.078	0.040	-0.075	0.001
Independent variables				
Dynamics	0.326**	0.199**	0.216**	0.304**
Mechanics	0.088	0.122	0.122	0.197**
Components	0.373**	0.369**	0.356**	0.294**
R^2	0.495	0.376	0.383	0.479
ΔR^2	0.480	0.370	0.372	0.478
ΔF	56.974**	35.531**	36.208**	54.965**

Note: N=187, **p<0.01, *p<0.05

and relatedness on brand equity and negatively moderates the impact of competence on brand equity. This paper not only enriches self-determination theory and gamification theory but also provides suggestions for gamification practice.

The findings of this paper show that most of the research hypotheses are supported by the data. Hypotheses H2b, H3b, and H4b are not supported, which means there is no significant effect of the game mechanics on autonomy, competence, and relatedness. One possible reason is that game mechanics typically set different game rules and conditions, which may impede the satisfaction of consumers' basic needs. For example, Pinduoduo's cash-out activity set the threshold by inviting the required number of friends. Many consumers cannot meet the standard and fail to withdraw the cash bonuses.

In Hypothesis H12, we supposed that openness positively moderates the positive impact of competence on brand equity. However, we got a reverse result: openness negatively moderates the positive impact of competence on brand equity. One possible explanation is that Individuals with open personality trait are curious about the internal and external world, open to new ideas, and willing to make independent judgements. It is hard to satisfy their competence because they set a high standard for the competence, thus weakening the positive relationship between competence and brand equity.

5.2 Theoretical Contribution

This paper makes several critical academic contributions. Firstly, it applies the gamification pyramid theory to explain the different effects of different gamification elements on brand equity. Game

Table 5: Hierarchical regression analysis of moderating effects

	Brand Equity		
	M1	M2	M3
Control variables			
Gender	-0.013	0.015	0.015
Age	-0.047	-0.045	-0.045
Education	0.009	0.006	0.006
Independent variables			
Autonomy	0.121**	0.114**	0.114**
Competence	0.375**	0.297**	0.293**
Relatedness	0.298**	0.198**	0.122*
Moderating variable			
Openness		0.412**	0.324**
Interaction items			
Autonomy *Openness			0.152*
Competence *Openness			-0.217*
Relatedness *Openness			0.156*
R^2	0.607	0.625	0.675
ΔR^2	0.130	0.018	0.050
ΔF	19.748**	6.324**	2.280*

Note: N=187, **p<0.01, *p<0.05

dynamics, mechanics, and components play different roles in gamification marketing. Secondly, it uses self-determination theory and the SOR model to examine how gamification affects brand equity. Although self-determination theory is most often used to explain gamification mechanisms, this paper enriches self-determination theory by using three basic psychological needs, namely autonomy, competence, and relatedness, as mediators. Third, this paper investigates the boundary conditions of the impacts of three basic needs on brand equity. Openness, one of the five core personalities, is found to moderate the effect between autonomy, competence, relatedness and brand equity.

5.3 Practical Implication

Enterprises should pay more attention to gamification, an important strategy to enhance brand equity. Specifically, the three dimensions of gamification differ in importance, with game dynamics and game component elements enhancing brand equity by positively influencing consumers' autonomy, competence, and relatedness. In contrast, game mechanics do not have a similar impact. Consequently, companies should devote their resources - human, material, and financial - in a focused manner.

Moreover, different personality traits of consumers, such as openness, will have different effects on the performance of gamification marketing. So, in practice, companies can design different gamification programs for consumers with different personality traits to improve the performance of gamification marketing.

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